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Bactrim_Dosing_FINAL

[00:00:00] Hello and welcome to Brig Points, the Society of Infectious Diseases pharmacist podcast. My name is Jillian Hayes and I am an infectious diseases and antimicrobial stewardship clinical pharmacist at Duke University Hospital in Durham, North Carolina. Today we are back with another episode in our dosing consult series.

This episode will join some fan favorites, episode 57, ceftriaxone, episode 61, And episode 80 dosing in obesity. If you haven't had time to listen to these episodes, I highly recommend all of them. I talk about them all the time while I am working through patient scenarios. These episodes are meant to be quick and represent a curbside question or thought.

I know I've certainly received pages about today's star medication, and I'm sure you have too. Of course, without further ado, we are talking about the [00:01:00] dosing of sulfamethoxazole trimethoprim. Now, I know that many folks have strong, and I mean strong, feelings about Bactrim dosing. Some folks may have strong feelings that I just referred to it as Bactrim, but if you listen to me try and work my way through sulfamethoxazole trimethoprim 50 times today, everyone's ears will be upset about that.

Recently, on Twitter, ID Doc Addy proposed that dosing Bactrim is like the doing your own taxes of medicine, and I think a lot of us spiritually agreed with that. Pharmacists chimed in all of a sudden and started likening themselves to accountants. Someone threw out algebra, the word chaotic was mentioned, and honestly seemed correct.

I think we can all relate to the sentiment. One of our own, Julie Justo, was quick to hop in the comments with a tweetorial that I immediately bookmarked along with 50 other people on Twitter. She does a great job of boiling down bactrim dosing into a four minute process. She says three minutes for concepts and one minute for math.

I'm not that fast, but even if I like quadruple the amount that I would take Julie, that's still only a seven minute process. And that's pretty good. We will link that thread in the show notes [00:02:00] for you all this to say, it's clear that we, the people are confused. So today we'll aim to ease some of that confusion.

By our discussion, we're going to talk about gram negatives, think gram negative bloodstream infections caused by enterobacterioles, methicillin resistant staphylococcus aureus, or MRSA, pneumocystis, and last, but oh, my friends, certainly not least, we will try to tackle stenotrophomonas, which the longer I'm an ID pharmacist seems to be the one pathogen that causes me to have existential crises on the regular.

So without further ado, let's introduce you to the stars of today's show, our panelists. Dr. Emily Heil is an associate professor in the Department of Practice Sciences and Health Outcomes Research at the University of Maryland School of Pharmacy. She's the pharmacy director of the Antimicrobial Stewardship Program and provides direct patient care on inpatient and outpatient ID teams.

She completed her undergraduate in PharmD at the University of North Carolina in Chapel Hill and subsequently completed her pharmacy practice and ID pharmacy residency training at UNC as [00:03:00] well. She's an associate editor for Clinical Infectious Diseases and is professionally active in the Infectious Diseases Society of America and the Society of Infectious Diseases Pharmacists.

Her research interests include individualization of antimicrobial dosing, particularly in critically ill patients. Antibiotic allergies, gram negative resistance, and antimicrobial stewardship on the domestic and international stages. Breakpoints Faithful are familiar with Emily. They love Emily. This is her third episode with us.

For more of her wonderful insights, feel free to tune in to episode 32 for an honest conversation on burnout and stewardship, which is still relevant today, and episode 60 for a review of important literature at last year's MAD ID meeting. Emily, welcome back. We are so thrilled to have you. Thanks so much, Jillian, for having me.

I'm excited to be here today and see you. Let's see if we can make sense of this topic. Me too. Next up, first time to break points, we have Dr. Andrew Frattoni, who is an Associate Director of Clinical and Laboratory Studies at the Center for Anti Infective Research and [00:04:00] Development, Hartford Hospital. He completed his undergraduate and PharmD studies at the University of Connecticut and went on to complete a PGY1 Pharmacy Residency at UConn Health.

Followed by a two year ID pharmacotherapy fellowship at Hartford hospital. He serves as a sub investigator on a multitude of phase one through four clinical trials and was recently a member of the CLSI ad hoc working group assessing breakpoints for *Stenotrophomonas*. This will come in particular handy for our chat today.

His research PD modeling, patient satisfaction, gram negative resistance, and PK in special populations. Welcome to breakpoints. Thank you so much for having me on Breakpoint. It's an honor to be here, especially with you two who are such seasoned Breakpoint veterans. This is certainly an early career highlight for me.

And I look forward to our discussion. I still feel the same about breakpoints. I call it the honor of my career, no matter how early on we are. Awesome. It's great to have you. Thanks so much again for joining us today. So I suspect [00:05:00] our listeners have at least some degree of familiarity with the medication sulfamethoxazole trimethoprim, but I like to start these dosing consult episodes getting everyone on the same page.

So to help frame our discussion today, Emily, can you kick us off with pertinent principles of sulfamethoxazole trimethoprim that we should keep top of mind? I'd love for you to briefly touch on things like mechanism of action, PKPD principles, and common indications just at a high level to get us started.

Sure. So today we're focused on Bactrim or whatever brand name you want to call it, a combination of two agents, trimethoprim and sulfamethoxazole. Sulfamethoxazole an interesting drug. It's been around since the thirties, but the combination of it with trimethoprim methamethaprim was FDA approved sometime in the 70s I think, but still it's one of these ancient drugs that we take for granted because we really don't know that much about it considering how long it's been around.

But the combination works by sequential inhibition of enzymes in the folic acid pathway. So your sulfa component is a dihydrotarate synthase inhibitor, and that's going to interfere with the enzyme. bacterial folic [00:06:00] acid synthesis by inhibiting dihydrofolic acid formation from para aminobenzoic acid or PABA, while the trimethoprim component is a dihydrofolic reductase inhibitor.

And so that's going to inhibit dihydrofolic acid reduction to tetrahydrofolate. So when you give this duo together, they are super synergistic and that gives them bactericidal activity against bacteria. a number of pathogens, including methicillin resistant Staph aureus and certain Enterobacter aureus.

It also has a super unique and frankly random spectrum of activity beyond those pathogens. So not only does it have bacterial activity, but it has some activity against certain fungi and bacteria. parasites including Cyclospora, Pneumocystis, Teleromycosis, which used to be known as Penicilliosis, and Toxoplasmosis.

So, I like to just think of Bactrim as a spectrum of just weird stuff. It comes as both an IV and a PO agent. The oral form is highly bioavailable, which is awesome from a dosing perspective. The [00:07:00] trimethoprim component's about 44 percent protein bound, and the self methoxazole component about 70 percent protein bound.

And they have relatively similar half life. life, so somewhere between 9 to 11 hours, which allows them to be given together in combination. Um, trimethoprim is really lipid soluble, has a huge volume of distribution, self methoxazole is a little bit lower, but together, you think about the drug really having great penetration of a lot of tissues, including about 40 percent CSF penetration, so it really gets to where it needs to go for most infections.

The combo does have pretty significant hepatic metabolism and both of the metabolites and about 50 percent of the unchanged drug are excreted in the urine. So this is one of those drugs that we think about having very high urinary drug concentrations, one of our fan favorites for things like pyelonephritis.

We also use it a lot for like MRSA coverage and skin and soft tissue infections and then some of the weird stuff that I previously mentioned. I think one big black hole in terms of its

PKPD, which I'm sure is going to come up a lot since we're here to talk about dosing, is what is the best [00:08:00] PKPD index to describe its activity?

And I'll be curious to see if Andrew has any thoughts on this, because he's probably dug into this way more than I have, but as far as I know, that's still a little bit of an unknown. And so this is not like a vancomycin where we have a clear cut PK PD index like AUC to MIC with a therapeutic drug range that we're targeting.

And so that can make some of this dosing that we're getting ready to dig into a little bit more challenging. Yeah. No, that's a great summation there by Emily. Not a lot for me to add. I know you want me to have all the answers on PK PD here, but I can't say that that's necessarily going to be the truth.

But I certainly appreciate you being the one who asked the question. Call out those really long enzyme names and the mechanism of action. So thank you for that. I tried. I was nervous That was the scariest part of this whole podcast. Yeah, hopefully that's a good sign that the scariest part is behind us We've used the phrase fan favorite weirdo stuff and we've already noted some known unknowns I like where this is going so far for our second and sort of last set the scene question.

I wanted to [00:09:00] ask, why is there so much turmoil around dosing of this drug? This is one of those medications that I see Twitter threads pop up from time to time, as well as just hear personal lamenting throughout clinical practice. Why do we seem to worry more and stew more over the dosing of this medication compared to other agents?

Andrew, I'll toss this one to you. Thanks Jillian. Uh, so I think there's a few reasons that dosing trim sulfa can be confusing or daunting. One reason is that dosing schemes for it vary so dramatically based on the indication in the pathogen. So we go from, you know, a few tablets a week to multiple tablets, multiple times a day.

But let me start, you know, with, with a positive aspect of dosing trim sulfa, because frankly, I don't think there are too many positives. But that is that the ratio of trimethoprim and sulfamethoxazole is fixed one to five for all drug formulations. And this as pharmacists, you know, makes our lives easier because we can focus on the dosing of one of the drug components and know that the other will always follow proportionally.

So in the case of trim sulfa [00:10:00] for dosing, Right. We focus only on the trimethoprim component, and this is critical in step one to dosing trim sulfa appropriately. As an aside here, for dosing combination agents, that's kind of one area in ID where it seems like we really can't help ourselves, but to complicate things, because there really is no standard, right?

So if you look at our BL BLIs, for example, uh, Sometimes we combine the BL BLI in the dose, like when you look at PIP TASO, sometimes we just ignore the BLI, like with the MoxClav, or sometimes we just split them into two separate parts, like with newly approved

Solder. So I think the take home point here is to always be on your toes when you're dosing combination agents in infectious diseases.

And with TrimSulfa, we're only referencing the trimethoprim for dosing purposes. And oftentimes this is expressed as a Mg per kg per day of trimethoprim. So for the oral tablets, uh, for TrimSulfa, we have single strength and double strength. And honestly, I find this type of terminology to be a little bit silly [00:11:00] sometimes, you know, it's kind of like, for some reason we wanted to make dosing TrimSulfa sound like it's close to an OTC pain medication.

Right. Where you have the regular strength and the extra strength, but really at any rate, what you really need to understand is that there's 80 and 160 milligrams of trimethoprim respectively in the single strength and the double strength tablets. And this equates to 400 and 800 milligrams of sulfamethoxazole when you apply that, that fixed one to five ratio, but the sulfamethoxazole is much less important to understand for dosing.

And so, you know, here's the part in the podcast episode, Jillian, where you're really going to want to edit in some sort of like dramatic music. I'm thinking like Phantom of the Opera, because I'm going to discuss about IV Trim Sulfa. So IV Trim Sulfa has to be diluted in D5W in every 80 milligrams of trimethoprim.

So think just one single strength tablet, right, at its most concentrated is going to deliver 75 mils of fluid. [00:12:00] And in some institutions, this can be as much as 125 mils. So, uh, the final concentration, right, of the compounded trim sulfur really does affect its stability, which is notoriously awful. It's two hours at the most concentrated at the 75 mils per 80 milligrams of trimethoprim.

And it's up to six hours at the least concentrated, which is 125 mils per 80 milligrams of trimethoprim. So when you really put this amount of volume into the context of a clinical dose, like two double strength tablets, you're looking at anywhere from 300 to 500 mils of D5 in free water volume. So when you're, when you're delivering that multiple times a day into critically ill patient populations, like with PJP or with steno, it's incredibly hard for these patients to tolerate that.

And it can cause some serious complications. So specifically, right, we need to be on the lookout for electrolyte abnormalities, Right, trimethoprim is very similar to our potassium sparing diuretics in that it [00:13:00] blockades the collecting tubule sodium channels, which can lead to hyperkalemia. And we also have to remember, like I spoke about earlier, that we're adding a lot of free water with IV trimsulfa, and this can certainly alter our other electrolytes like sodium.

Certainly, trimsulfa also commonly causes GI related adverse events like nausea and vomiting. And these adverse events that I've mentioned here tend to be dose dependent. So it's important to make sure that we're not unnecessarily overexposing our patients to trim sulfa. I will just add that the IV formulation fluid burden and stability issues is no joke.

And I recall once having a patient where due to their body weight and the infection that I needed to treat them, we actually couldn't find a dose where we could give the patient the drug before it would reach its stability kind of time point. We were like running. that drug from the pharmacy to the ICU and it's still like almost clocked out from a stability standpoint before we could get the whole dose in.

It's just [00:14:00] crazy. So thank goodness for good bioavailability, right? Amen. Any drug that involves sprinting, I think I'm out immediately. Absolutely not. Jogging should not be involved in medication administration, but we'll do it if we have to. Okay, I think that that was a very helpful summary of why, one, this is 100 percent complicated, right?

So feeling overwhelmed by this topic is understandable. We'll get into the fact that answers are hard or scarce, and there's a lot of concerns that we need to look out for. So, anything else to add before we dive into pathogen specific discussions? I think we could probably talk about potential Bactrim related toxicities in a whole other podcast, so I think good to go into dosing from my perspective.

That's fair. We'll cut ourselves off lest we be here all day. All right. Fair enough. So now that we've set the scene, like I said, we'll dive in sort of one pathogen at a time. So we're going to begin talking about sulfamethoxazole trimethoprim dosing in the setting of infections caused by common gram negatives.

So, gram [00:15:00] negative bloodstream infection caused by enterobacterioles is sort of the, the umbrella scenario that we'll talk about here. Emily, you were the primary author on one of my most favorite papers. Of all time, a consensus guidance document called optimizing the management of uncomplicated gram negative bloodstream infections.

I'm a complete evangelist for this document. It's been a super helpful reference for so, so many. So first off, thank you sincerely to you and the author team for creating this in the document. For patients with uncomplicated gram negative bacteremia, oral antimicrobials can be used with fluoroquinolones and Bactrim representing our preferred options.

The dose recommended in this document is five milligrams per kilogram every 12 hours, which works out to about two double strengths tablets every 12 hours for a 70 kilogram patient. How did y'all come to this recommendation and what should we know about Bactrim dosing in this clinical scenario? Yeah.

So this paper was a lot of fun. So we had six ID physicians and six ID pharmacists that were participating in this panel and the senior author on the paper, Pranita Tama, [00:16:00] she referred to that table with the dosing as pharmacists gone wild. She was so entertained by how obsessed she was. and how down, deeply down rabbit holes we went to put that table together, but you know, nothing like pharmacists and being anal and wanting to be precise on our dosing as much as we can.

And I will say, with great sadness, since this is the topic that we're kicking this off with, that we're probably starting with the one that actually has the least amount of data to go into it, so

my answer on this might not be as satisfying as everyone was hoping for. But, You know, again, as a reminder, Bactrim came out in the 70s and we don't actually have a great PK PD index to target.

And so it's not quite as simple, not that any of this is simple, but, you know, with some drugs where we at least have kind of a defined PK PD target and you can go back and dive into your early PK literature, you can kind of piecemeal something together, but not quite the case with Bactrim. So what we did, we went to KUSERS.

Everyone's familiar with KUSERS. favorite textbook of, kind of, I. D. Bible about our, our antibiotics. And we [00:17:00] looked at any literature that was cited within Kusser's. We went down some rabbit holes looking at, again, 1970s era, uh, pharmacokinetic literature. And then the other thing we did was, there's a lot of papers that have come out in probably the last 10 to 15 years that have evaluated using Bactrim as an oral step down option for gram negative bloodstream infections.

And so looking at some of those retrospective reviews and seeing what dose the authors used in those studies and pulled that together and really it reflected what is in the package labeling for Bactrim as an the go to dose for what's considered to be like a moderate infection of somewhere between 8 to 12 milligrams per kilo per day divided 2 to 3 times per day.

And so we, we went with 5 milligrams per kilo every 12 hours, just based on that information. I can't say that this is the most solid database recommendation of the ones that we made in that paper. And frankly, I think that the biggest issue is just that lack of PK PD index to target, but from what we were gathering, at least from [00:18:00] other places that have used it consistently as part of their step down therapy for gram negative bloodstream infections, that seems to be consistently what people are using in practice.

And so that plus the PK literature is where we landed. It's awesome to always hear the history about how certain documents get created and kind of what really goes into it, the real truth. And sometimes, you know, this is the best that we can do. So certainly commend, you know, all of the efforts of all involved.

Also, I would like to note that if this episode did not already have a title, as it is part of a series, Pharmacists Gone Wild would most certainly be in the running for the title of this episode, for sure. Okay, easy enough. This is just a great, it was a great place to start. We've eased into the conversation.

Okay. And then from here, we're going to move into MRSA next. So we know that sulfamethoxazole trimethaprim is one of our only oral antimicrobials with activity against MRSA. So this is an important conversation. Andrew, I'll toss this one to you. Where should we start? What [00:19:00] should we think about when we are using Bactrim for MRSA infections?

All right, here we go. So, you know, I certainly agree that TrimSulfa is an important oral option in our toolbox for MRSA infections, especially for skin and soft tissue infections. So if we look at the 2014 IDSA SSTI guidelines, they do list TrimSulfa as a recommended agent for purulent infections, where MRSA is a likely pathogen.

So in here, the recommended dosing is one to two double strength tablets twice a day. So I think the question always remains. So who is that one double strength tablet twice a day? And who is that two double strength tablet twice a day? So, you know, luckily for us, there have been a few studies that have evaluated this question.

There was a study by Kadena et al. That was published in 2011 in AAC. This was a prospective single center study that evaluated one versus two double strength tablets twice daily for the treatment of MRSA SSTIs for a treatment period of 7 to 15 [00:20:00] days. And they found no difference in the clinical resolution between the two groups.

Looking at another study in 2012, I was retrospective and looked at 106 hospitalized patients who were treated for cellulitis. This was by Halilovich and colleagues. They found that morbidly obese patients who were prescribed the one double strength tablet twice a day experienced greater clinical failure than those who got the two double strength tablets twice a day.

So, you know, with this limited amount of evidence, I think the dosing takeaway for TRMSulfa and MRSA skin and skin structure infections is that. For the most part, one double strength tablet twice a day is probably going to be appropriate for most of our patients, but there are certain patient characteristics, I think, that may warrant the two double strength tablets twice a day, such as weight succeeding 100 kilograms or immunosuppression.

Again, not a ton of data to make this call per se, but in general, I think, you know, trying to target around five mg per kg per day of trimethoprim is a reasonable [00:21:00] approach. Uh, for MRSA SST. I like that list, Andrew. I practice the same way. I err on the one Q12 just for better tolerability. And I agree with you.

Weight is going to be my main driver of when I'll pull the trigger on doing two gram, uh, two double strength tablets twice a day as well. The only other thing I would add to that list. Next is also trauma induced skin and soft tissue infections. I think that was one of the demographics in that first study you mentioned where the baseline groups just weren't quite the same and there may have been an edge for the two double strength versus one, but otherwise, I'm right there with you on that.

Yeah, thanks, Emily, for that addition. I think that's a great piece to add. Now, moving beyond skin and skin structure, a little bit deeper, uh, into the, the physiology, I think Tremsulfa certainly has some merit in the treatment of bone and joint infections that are also caused by MRSA. Now, as Emily alluded to in the introduction, All right, trimethoprim has a pretty high volume of distribution, especially relative to sulfamethoxazole, and it enters the bone with about half [00:22:00] of the exposure that you see in serum, while sulfamethoxazole is a little bit lower at maybe 15%.

And trimethoprim has been shown to get into the synovial fluid with pretty similar exposures to that of serum. So with that being said, You know, I think it's important to first decrease the bacterial burden of some of these infections, whether it's surgically or with other MRSA active agents, you know, especially if there are abscesses or there's a lot of necrotic tissue involved before we really think about using trim sulfa as monotherapy, uh, in my opinion, and this is due, um, in large part to, to exogenous thymidine that can be found in the body.

And thymidine really acts as a salvage pathway for folate synthesis. So if you're using, you know, trimsulfa for an active treatment of bone or joint infection, I really want to make sure that, you know, we have our, our source well debrided. Um, and I would suggest probably targeting closer to 10 mg per kg per day of trimethoprim really to just help account for some losses of penetration.

[00:23:00] Um, and then Uh, in the presence of exogenous thymidine. So not a ton of data yet again to, to make this, um, personal preference of dosing recommendation, but that's where I generally fall. Emily, anything to add? Nothing. I agree with Andrew's expert opinion there. I think that's a great way to approach it.

Awesome. I was secretly hoping the phrase exogenous thymidine would come up on this podcast, so thanks for adding that into our bingo card today. happy to oblige. All right, so where you really start to lose me for using uh, Bactrim for MRSA infections is in the space of bacteremias. So even when bacteremias appear fairly transient, like a line infection, you know, and the line has been removed, I just have too much respect for MRSA and staph in general in the bloodstream and how sticky it is.

So, you know, I worry that it has traveled elsewhere and created a pocket of infection that is just rich with thymidine. So personally, I really don't advocate for trim sulfa in [00:24:00] MRSA bacteremia. And additionally, the clinical evidence really isn't very supportive of it either. I know there's one study in the British Medical Journal in 2015 that actually was a randomized control trial of trimethoprimsulfa versus vanco for serious MRSA infections.

And they found that trimsulfa was not non inferior. to vancomycin. And that was especially true amongst the patients who had bacteremia in that cohort. And so personally, I'm not even the biggest fan of vancomycin's efficacy in MRSA bacteremia. So if agents can't go toe to toe with vancomycin, I'm generally not too impressed.

So ultimately, I don't routinely, you know, endorse or use trimsulfa for the treatment of MRSA bacteremias. I really appreciate how you put it that you just have, it's not disrespectful to Bactrim, right? You just have too much respect for the pathogen. And I think that's a helpful perspective to adopt.

We're not hating on the drug. We just respect the pathogen too much to put the drug in this position, right? We're [00:25:00] asking Bactrim to do a whole heck of a lot there. Exactly. That's a, that's a heavy carry. And taking a shot at vancomycin, I know some of our listeners will be really pleased to hear that as well.

So, man, if you can't even, like you said, go toe to toe with vancomycin, should we be doing it? Uh, before we transition to the next pathogen, Emily, anything to add about MRSA bloodstream infections? No, that's such a complicated question, and I think I think, Andrew, you summed it up really well. I know when we've gotten in some real pickles with some, you know, super tough cases with some challenging social situations and you're kind of pinned to the wall, it has come up for discussion.

And I remember pulling once this paper from, I think, the early 90s that evaluated oral bacterium versus vancomycin for MRSA endocarditis in, um, persons who injected drugs and had limited treatment options. Bactrim was inferior to VENC in that study, but if you look at the overall clinical success rates, I want to say they were in the 70 ish percent range.

So it wasn't a complete abysmal failure, but we can do better in most situations. And so I think because [00:26:00] we have better options, most of them IV, but you know, certainly we could debate maybe Linazolid being more of a shining star here. I think that I'm with Andrew, and if you can't be better than VENC, then oof, let's let Bactrim shine where it shines, which is not here.

Perfect. I love that. Yes, we can do better in most cases. I think that's the perfect way to summarize it because absolutely in those cases where social situations or what have you are going to necessitate the use of an oral antimicrobial, something, a 70 percent success rate is better than absolutely nothing.

I think everyone would agree there, but yeah, let's let it live in its spotlight and tell it, you know, this isn't where you shine. And that's, that's our job, right? I think if we, uh, viewed our job as pharmacists as just helping the antibiotics shine where they shine, that's a, a different perspective that helps us advocate well for them.

Okay, so just like that, we are halfway through. I like to think I've saved the most torturous topics for last, so we're really building to a, a big finale here. We will tackle pneumocystis first. [00:27:00] So, starting with prophylaxis first and then we'll talk about treatment dosing. So, if we're looking at the guidelines for prevention and treatment.

of opportunistic infections in adults and adolescents with HIV. The guidelines state that one double strength tablet daily is the preferred regimen, but one single strength tablet daily is also effective, might be better tolerated, and then one double strength tablet three times weekly is also effective.

I think a lot of our listeners, and I know speaking from personal experience, I've seen some creative bacterium prophylaxis strategies. My favorite that I've ever seen is just twice daily on Monday, and then we'll see you again next Monday. What do we do with this? How do we make sense of these, these different recommendations?

This is such a fun topic because I think one of the coolest things about it is that all of our data for how to dose Bactrim for this comes from the early 90s and since then we just have come so far as it relates to effective antiretroviral therapy and it makes it really hard to to go

back and study this question because we will hopefully never [00:28:00] again see the population numbers from those studies in the early 90s to actually get better data on how to optimally dose a spectrum.

So that's cool. I'd rather have no data and live in a world where HIV is a manageable disease any day. That being said, um, really when we're thinking about Bactrim for PCP prophylaxis, it's weighing that fine balance of optimal tolerability with efficacy. And so some of the early studies where it was evaluated primarily for secondary prophylaxis of PCP, PJP, whatever you would like to call it, um, the dosing was one double strength tablet one to two times per day.

And so in the first major another study that evaluated Bactrim for PJP prophylaxis. They actually use one double strength tablet twice a day. It was an open label study comparing it to dapsone and aerosolized pentamidine, but what was interesting was that because it was open label, it did allow for dose adjustments and only 21 percent of patients in the Bactrim arm ended the study on their originally assigned dose.

And so it became very evident that [00:29:00] one double strength tablet twice a day was too much for people to tolerate. It wasn't going to be a great option even though it was the most effective. And so from there, there were some subsequent studies around 1995 up to 1999. It started to look at different regimens to evaluate for tolerability.

So there was a 1995 Dutch study that looked at one double strength daily versus a single strength daily for prophylaxis in patients with a CD4 of less than 200. And they found no difference in PCP rates overall, but less adverse drug events in the single strength group advocating for potentially just a single strength for all.

And then after that, there was another randomized controlled trial that looked at a double strength daily versus just a double strength three times a week. Again, found Um, relatively favorable rates of efficacy, um, in terms of PCP prevention, but better rates of tolerance in the patients that only got it three times a week.

So, the take home of the guidelines is that the double strength has the most long standing data in terms of its efficacy, but the [00:30:00] alternate regimens of just a single strength daily or a double strength three times a week might actually be better tolerated. Bactrim is just light years better than our alternate prophylaxis.

Um, some of our physicians that we work with used to joke around, like, if a patient's actually taking their Bactrim prophylaxis, they're not going to get PCP, whereas with Dapsone or Atovaquone, it, uh, could happen. So, um, you really want to keep patients on Bactrim. And so, if it means switching to the single strength or just doing the three times a week dosing to improve tolerability, then that's fine.

But one of the frequently cited reasons to prefer the double strength tablet over the single strength or three times a week. Dosing is if you're also trying to um, confer cross protection for toxoplasmosis. And so a frequently cited reason to use a double strength Bactrim is that

the double strength dosing is what has been shown to best confer protection against toxoplasmosis, and that comes from a 1992 study, but interestingly that study evaluated dosing of just two double strength tablets given two times per week as primary prophylaxis.

So it's [00:31:00] interesting that that's still what we kind of reference back as to why we prefer the two double strength tablet, or excuse me, just double strength dosing over single strength dosing, um, even though it wasn't actually one double strength every single day. So I still think that that is a little bit of a murky, uh, subject, and my kind of personal take home is the best dose of Bactrim is the one that the patient can tolerate, and so if that's just a single strength daily, then I think that that is the best.

The best way to go. I love what you said about the best dose is the one that the patient's going to take. I think that's true really across the board, but especially in this instance, given the power that Bactrim has for prophylaxis in this space, absolutely. Whatever the patient can tolerate, sounds good to me.

And I really just want to point out that PJP is a fungal infection. And here we are using, you know, this antibiotic. So that, that is always cool to me. Absolutely. Shout out to backroom, right? See, this is a place where I can shine. This is where it gets the spotlight and deserves it. Okay, perfect. So now that we've sort of made sense of.

Prophylaxis, we'll [00:32:00] move on to treatment. Our option for moderate to severe PJP is 15 to 20 milligrams per kilogram of trimethoprim given every 6 or 8 hours. And then mild to moderate, it's going to stay the same dose, just has the recommendation to divide every 8 hours. For mild to moderate PJP, there's also an option to pursue two double strength tablets three times daily.

The guidelines bring up the word tolerability for prophylaxis, we just talked about it. So I know it can be a concern here too. These are hefty doses of trimethoprim, so what guidance do you have for our listeners? Yeah, Bactrim dosing for PJP treatment is so poorly tolerated. It's interesting in the guidelines, they actually cite that treatment failure attributed to dose limiting toxicities with Bactrim occurs into up to one third of patients, which is just crazy to think about.

So yeah, this is a very common issue that we have to deal with, but again, I'm going to go over some data and then I'm going to say we don't have any [00:33:00] more clinical data and I don't know that we're going to be able to get it because this is just not that common of a disease state anymore. which is the best news.

Um, but interestingly, if you dig way back into the original literature on this subject, the adult dosing for back term for PJP pneumonia was actually extrapolated from pediatric cancer patients because that's where the disease state was most common at the time. So believe it or not, when we first started treating patients with HIV that had PJP, we were using doses of greater than 20 mg per kg of trimethoprim divided.

shockingly, that was very poorly tolerated. So, in the mid 90s, there was a lot of attempts to try to optimize the dosing of Bactrim for this indication because of the tolerability concerns. So, the original dosing that was used in the Bactrim Most common clinical studies was the 15 mg per kilo divided every 6 to 8 hours.

And then there was actually some subsequent clinical trial, or sorry, PK data that came out that suggested an [00:34:00] alternative dosing of given a 15 to 20 mg per kilo trimethoprim component kind of load over your first 24 hours, divided up of course, followed then by a 12 mg dose. Milligram per kilo per day divided dose for the remainder of therapy that was only demonstrated to be equally efficacious in a, in an in vitro model.

But all of the clinical data that predated that, still used that 15 to 20 milligram per kilo per day dose. So the guidelines unfortunately still don't recommend lower doses because the clinical efficacy of these dosage regimens haven't been confirmed, even though there are some PK data suggesting we can go lower.

There are 2 studies that start to point in the direction that maybe we could go a little bit lower. There's a 2009 retrospective review from Auckland City Hospital where patients that were treated with a dose of trimethoprim 10 mg per kilo per day divided had comparable efficacy to treatment that was with higher doses in previously published studies.

So this was a non comparative review. They just described their experience with [00:35:00] patients that got the lower dose at their institution and looked at outcomes data historically in the literature. And demonstrate that patients in their cohort seem to have similar mortality rates compared to patients in previously published literature.

So, obviously limited single center retrospective study, but at least some relatively modern data, not 30 years old, that suggests this lower dose might be okay. And then most recently in 2020, there was a systematic review published in OFID that included six studies that evaluated lower doses of Bactrim, um, so all of these studies included doses of less than 10 milligrams per kilo per day of the trimethoprim component and found similar rates of mortality but significantly fewer treatment related adverse drug events compared to standard doses of 15 to 20 milligrams per kilo per day.

But, unfortunately, all six of these studies, again, were observational in nature. They included that, um, Auckland study that I just mentioned. And so we don't have a lot of modern clinical data that validates these lower doses that were identified in the in vitro study. [00:36:00] So personally, I still come out kind of guns blazing with the initial 15 milligram per kilo dose.

Um, and then if the patient's having issues tolerating it, I will potentially pull back and try a small dose decrease before switching entirely to another regimen. Right now, the guidelines Do say if patients are having, uh, treatment related adverse drug events instead of just adjusting the dose to just go ahead and switch all the way to another regimen.

But I think it's reasonable now that we are starting to get a little bit of data that lower doses might be okay from an efficacy standpoint to try that before you switch altogether. The other

thing that I think is completely fine to do, I end up doing lots of funky splits of the dosing. So, you know, if you.

math out your dose that the patient needs five tablets per day or seven tablets per day, you know, split it up, maybe two tablets morning and dinner and one at lunch. Um, I've come up with some very funky ways to try to spread out the backroom throughout the day. Time it with meals to try to help with tolerability a little bit, but it can be a challenge.

Yeah, I really like the, uh, guns blazing, uh, analogy there, but, you know, I, I like the [00:37:00] ability to, to try to be creative and I think that's a, a good strategy. Absolutely. Okay. So I think to summarize, all gas, no brakes to start, tap the brakes if you have to. Get creative in the interim. Thank goodness this disease is not as common as it used to be.

Okay. My friends, the time has come. I just took a big sigh. You could probably hear that on the microphone. I mentioned that this organism induces existential crises inside of my heart. Its name is *Stenotrophomonas*. This whole situation has only gotten more complicated recently with the recent comment from CLSI and the update to the IDSA gram negative guidance document to only use trimethoprim sulfamethoxazole as a component of combination therapy at least until clinical improvement is observed.

To briefly review current dosing recommendations at least for all infections other than cystitis, the guidance document suggests 8 12 mg per kg per day of the trimethoprim component divided every 8 12 hours. [00:38:00] With a consideration of a maximum dose at 960 milligrams of trimethoprim per day. Andrew, this is a heck of a welcome to BreakPoints, okay?

We're asking you to come on and just casually talk about bactrim dosing for steno, but let's do it. We'll come to you first. You and your team have done a ton of work in this space. Is everything with steno made up as I suspect? What do we know for sure? Help us out. As much as I'd love to save you from your existential crisis, Jillian, it's possible that I might make this even worse for you.

So, is everything for Steno made up? Um, I don't know if I would say made up, but I will certainly say that the majority of its historic breakpoints have been borrowed from other species, with the assumption that they're going to apply to Steno as well. And a main issue with that is that there's been some more modern PK PD evaluations on the breakpoints for certain bug drug combinations that have been updated.

So if you look, for example, fluoroquinolones [00:39:00] for antierobacterials and *Pseudomonas aeruginosa*, but the steno breakpoints, which were essentially inherited from these older breakpoints of these other bacterial species that have since been updated, are now remaining the same. And there's a, you know, a real PD data to try to support or refute, you know, these existing breakpoints.

So thanks to the FDA for providing some funding streams for more modernized assessment of PK PD for some older drugs. And certainly thanks, uh, you know, to my mentors here in

the lab, Dave Nicolau and Joe Cuddy for proposing and carrying out, you know, some of these assessments against Steno, one of which importantly for this podcast was Trim Sulpha.

So as I've alluded to previously, right, thymidine is an endogenous salvage pathway, right, that bacteria can use against dihydrofolate reductase inhibitors like Bactrim. But unfortunately for us PKPD people, rodents have multitudes high level of thymidine relative to [00:40:00] humans, um, and rabbits and monkeys.

And so this makes our gold standard murine PKPD infection models, uh, much less translational. And so therefore, you know, our lab at Hartford used a couple of in vitro models to assess trim sulfa PKPD against steno. Uh, and these findings are published in a 2022 AAC paper, and then subsequently in a 2022 JAC paper by my co fellow Maxwell Lasko.

So the main findings from these studies was that 20 mg per kg per day of trimethoprim, so, you know, guns blazing, all gas, no brakes type dosing, um, that kind of exposure could not produce even stasis in a 24 hour time kill analysis, even down to an MIC of 0.5. But interestingly, and cleverly, when you put an E.

coli in that same exact model that has the same exact MIC, you see multi log kill. all the way up to an MIC of two, which is the break point for both E. coli [00:41:00] and steno against Bactrim. Uh, but as, as the saying goes, right, all models are wrong, but some are useful. So, time kill analyses, you know, they expose the isolate to a fixed antibiotic concentration, and we know that clinically, concentrations rise and fall, right, over a dosing interval.

So next, they, they took a one compartment in vitro chemostat model, which clears and add drug into the inoculated reservoirs, and again, simulating these really high AUC exposures that are, are relevant to 20 mg per kg per day of trimethoprim. It's producing stasis, you know, but no net killing, and even when they increased the exposure to what would be equivalent of 100 mg per kg per day of trimethoprim.

Which is just crazy, and no one would ever do, the model still wasn't able to even get a one log reduction in bacterial burden. And so, using modeling based on this one compartment in vitro chemostat experiment, yielded a [00:42:00] free AUC to MIC target of 67. So I know Emily was looking for some targets earlier.

So here's one free AUC to MIC target of 67. Um, but I will say we weren't able to do, you know, kind of the classic dose fractionation to pull apart which PK PD index, you know, is, is most predictive of change in bacterial burden because there was really just no dose response. Um, you got to stasis and that was about it.

So usually when you, when you can't find a, a really resolute PK PD target, you just kind of kick the can down to AUC over MIC. So here they found a free AUC to MIC target of 67. And this was then used in an unpublished Monte Carlo simulation based POPPK in the literature. Uh, and to reach a 90 percent probability of achieving this free AUC to MIC stasis target, the MIC had to be.

0. 5 or less using simulated trimethoprim doses of 5 mg per kg Q12 [00:43:00] and even increasing that dose of trimethoprim to 15 mg per kg per day really did not appreciably raise the the probability of target attainment. So pushing the dose at least according to this model does not appear to give you you know better bang for your buck and really only able to to reach a stasis target which in the PK PD world is not ideal you know for serious infections.

And there's been other in vitro studies that have shown similar findings without doing the formal PK PD modeling. There's been time kill studies by way at all in 2016. There's been another one compartment in vitro model by Zelenetsky et al in DMID in 2005 that showed that Bactrim monotherapy over 24 and 48 hours led to growth.

But when used in combination with other agents did leave to, to multi log kills in, in many situations. So, you know, I said that rodent models are not. translational for Bactrim. However, there [00:44:00] has been one study that has used TrimSulfa in a rabbit pneumonia model. This was done by the folks in Cornell in an AAC 2022 paper where they administered 5 mg per kg Q12 of trimethoprim.

Uh, to the rabbits, but there were no PK studies, uh, to understand what that exposure looks like and how it correlates to humans. So who knows what 5 mg per kg Q12 of trimethoprim in a rabbit looks like? I have no idea. And then just, you know, kind of the the cherry on top is that unfortunately the clinical data is extremely confounded with steno and bactrim, you know, Kind of the old adage of do they die from steno, or do they die with steno?

The existing data tend to all be retrospective studies. They commonly don't report MICs And TRMS alpha is usually given in combination with other agents in these, you know, retrospective studies. So really as it stands, I think there's insufficient clinical PK [00:45:00] PD and micro data to really suggest that You You know, when you see an S on A.

S. T. results for steno that it's really going to stand for clinical success when it's used in monotherapy. And, you know, the problem here is that other drugs that we test for steno also tend to be less than great. So, you know, it really is a really tough treatment landscape. And I think there's no clear options.

You're correct. My existential crisis has gotten a bit worse, but that was an incredibly helpful summary. So thank you. And I feel that that ominous dark music that you asked for, I believe Phantom of the Opera was the request. We should have, um, intro'd that segment with that music as well. Uh, Emily, what can you add to this?

Very sad discussion about Bactrim dosing in Steno. Wow, no, I'm still just processing. That was some great information and some great insights there, Andrew. I mean, I think to me, Stenotrophomonas has always sounded like it should have been a dinosaur, and I do kind of equate it to a roach, which has been around, they've been around since [00:46:00] the dinosaurs were around, and sometimes I feel like the more we learn about, That bug, the less we know, and yeah, I think my main take home any time I'm dealing with steno is just to take a lot of the information I'm getting from my lab with a grain of salt, not Any issue with the

lab and what they're doing, but just that there's so much that we don't know about how to optimally treat this pathogen, and that was really insightful information you shared.

Yeah, I mean, I think the biggest piece of advice with steno is just, you know, is it a colonizer? Is it just there for the ride, or do we truly think it's causing some sort of clinical infection? Because, you know, otherwise I feel like a lot of these steno drugs get used more as an anxiolytic. Uh, for us as clinicians, as opposed to really, you know, helping treat certain patients supposed infections.

We talked a little bit about the, um, anxiolytic powers of ceftriaxone on the emergency department stewardship podcast. And, uh, as [00:47:00] much as they might make us feel better, these therapies are too limited and probably too intolerable for patients to be giving them out willy nilly for our own anxiety when it comes to steno.

Okay, man, that was helpful and we needed to hear it. Thanks for the tough love there with Bactrim and Steno dosing. Um, I want to give you guys one sort of last chance question. Is there anything that we haven't covered? I know there's loads we could still talk about, but anything that you, we haven't covered that you want to mention or anything that you want folks to keep sort of top of mind the next time that they are making dosing recommendations for sulfamethoxazole trimethoprim.

Okay. I think the main things to keep in mind are just to be careful with how you're doing your math and remember that the dosing is based on the trimethoprim component. As Andrew was going over earlier in the podcast, it is a little bit weird. We aren't consistent in how we approach a lot of these combo therapies and referring to the dosing, but just be really careful with your language and, and how you're [00:48:00] making your dose recommendations.

I try, if I'm ever asked for a dose, to just go ahead and get the patient's weight and stats and give them an actual dose and not a weight based recommendation because there is so much error potential there. Um, and, you know, it's by no means a benign drug. We could have done a whole pod on bactrim related adverse drug events.

And so, lowest effective dose possible that we can get away with I think is, is the key. And as I said for the, the PCP section, I am optimistic that maybe one day we'll have some data saying we can go lower than we have been. been, but we don't need to be shy about meeting the patients where they are. If they cannot tolerate the drugs at the dose, we've given them.

I got nothing to add. That was great. Okay. We've made it last, but most certainly not least we're ending on a high note. Officially, uh, we'll pivot to our segment called I feel nerdy. I feel nerdy is meant to be a safe place for our panelists to nerd out over their favorite ID topics, quirks, and fun facts for today's edition.

I would love to know [00:49:00] your favorite name. That we call sulfamethoxazole trimethoprim. Uh, there are many to choose from. It can be an Abrev, it can be a brand name. Uh, would love to know your favorite international or otherwise name for this

wonderful drug and why. So where I hail from in the South, the first brand name that I heard of for Bactrim when I was working at a small independent community pharmacy during pharmacy school was Ceptra.

We actually put all of our antibiotics on the shelf alphabetically and, and backdrome was in the S's for Sceptra, not Sulfamethoxazole, but Sceptra, because we had them organized by brand name. Really helped me with my Top 200 back in pharmacy school. Um, and that's all I knew that drug is. And so when I moved up to Maryland, which some people say the South, is the South still, but I say it's not the South still.

So I moved up to Maryland and I'm referring to this agent I'm Team Sceptra, and people are looking at me like I have two heads, and I have never ever heard that brand name used outside of the South, and I don't know why that is, but I'm Team Sceptra just because it gives me, like, good home [00:50:00] vibes and reminds me of that good old days in that independent pharmacy.

I like it, Emily, and home vibes are always good. Uh, but in full disclosure, I am a Bactrim guy. But I recently discovered that there's a foreign branded version called Megatron. And for me, that absolutely takes the trophy, right? So not only does it sound pretty tough, uh, but that brings me back some nostalgia because it sounds a lot like a Megatron.

So it kind of reminds me of, you know, watching Transformers growing up with Megatron and the Decepticon cronies kind of taken on Optimus Prime. So. That's that's where we've come. Megatron. I like that. I feel like I'm going to have to break that out on rounds the next time I'm recommending it. Like we got to bring in the Megatron guys.

That's a good one. That'll get you a couple looks for sure. Absolutely. Thank you for discovering that because I had not heard that prior to this moment. So Megatron, we will have to weasel that into a note or two for sure. Um, [00:51:00] I was going to go with Co Trimoxazole, it's a little international, it gives it a little flair, I kind of like it, um, but in terms of, yeah, day to day, I think I say Bactrim the most out of, out of anything for sure.

Alright, thanks for humoring me, um, and thank you both so, so much for helping us unpack. This topic that is by no means simple, by no means straightforward, um, and sharing your clinical opinions. It has helped me learn so much, I feel like, in the past hour, and I'm sure our listeners will feel the same. So thank you both for joining us today.

Thanks, Jillian. It was fun. Thank you, Andrew. I learned a lot from you. Thanks for having me. You know, it's a pleasure to finally join the BreakPoints family here. And thank you, also, our listeners, for listening to BreakPoints, the Society of Infectious Diseases Pharmacist podcast. I've been your host, Jillian Hayes, and our featured speakers have been Emily Heil and Andrew Fraton.

BreakPoints was created by Julie Ann Gesto, Aaron McCreary, and Jason Pogue. This episode was produced by myself and Jeanette Bouchard. It was edited [00:52:00] by Nick

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